HOCHSCHULE ESSLINGEN

APPLICATION PROJECT ON EARNINGS MOMENTUM BACKTESTING

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TABLE OF CONTENTS

..... 1
TABLE OF CONTENTS..... 2
1. Hint / Initial idea.....3
1.1 Initial observation and motivation.....3
1.2 Earnings announcements and price reaction.....4
1.3 Identification of post-earnings momentum bias.....6
1.4 Market-wide test: Is there a universal post-earnings momentum effect?.....7
1.5 Portfolio building of the momentum strategy.....8
2. Analysis and selection of performance metrics.....9
2.1 Trading strategy definition.....9
2.2 Portfolio construction and return computation.....10
2.3 Selection of performance metrics.....12
3. Results analysis..... 14
3.1 Equity curve and long-term performance..... 14
3.2 Forward-looking performance and sensitivity to entry timing..... 18
3.3 Distribution of monthly returns.....21
3.4 Distribution of yearly returns.....23
4.Pipeline execution.....25
5. Conclusions, risks, and potential improvements.....26
References.....28

1. Hint / Initial idea

1.1 Initial observation and motivation

The initial motivation for this project arose from a personal idea after the direct observation of market behavior during 2024, rather than from a predefined theoretical model or an attempt to optimize an existing investment strategy. After several months of strong equity market performance, market conditions began to shift toward a more uncertain and volatile regime.

Throughout 2024, increasing macroeconomic and geopolitical uncertainty became evident in asset prices. Factors such as inflation dynamics, monetary policy uncertainty, and global political tensions contributed to a less predictable market environment. This instability intensified around mid-2024 after a strong bull market and became particularly visible in the months around August 2024, when the reversal of the Japanese yen carry trade triggered a sharp equity market sell-off.

During this period, earnings announcements stood out as moments of unusually strong price movements. While earnings events are always associated with elevated volatility, the magnitude and frequency of price reactions observed in 2024 appeared significantly larger than in more stable market phases. Many stocks experienced abrupt and pronounced movements around earnings, often exceeding what would be expected based on historical behavior.

These observations suggested that earnings announcements were acting as focal points where firm-specific information interacted with broader macroeconomic uncertainty, amplifying price volatility. Importantly, this heightened volatility was not confined to a small group of speculative stocks, but was observable across different sectors.

A key insight emerging from this experience was that, although the direction of price movements around earnings announcements is often difficult to anticipate, the presence of volatility itself appears to be a persistent feature. This led to a shift in perspective: rather than attempting to predict the direction of post-earnings price movements, it seemed more relevant to explore whether a strategy could systematically benefit from the recurring volatility associated with these events.

This intuition formed the starting point of the project and motivated a deeper analysis of earnings-related price behavior, with the aim of assessing whether these observations could be translated into a coherent and systematic investment strategy.

1.2 Earnings announcements and price reaction

Earnings announcement dates were obtained using a custom Python library developed for this project, named *stocks-earnings-dates*. The library contains historical earnings announcement dates for all S&P 500 constituents since 2003 and was constructed by programmatically extracting and parsing corporate filings from the U.S. Securities and Exchange Commission's EDGAR database.

In order to analyze how earnings announcements affect stock prices, earnings-related price movements were decomposed into distinct return components. This decomposition allows for a more precise characterization of how information released during earnings announcements is incorporated into prices over time, rather than treating earnings-day movements as a single homogeneous event.

Three measures of price reaction were considered. First, the overnight return from the close of the earnings day—after the earnings are announced—to the opening price of the following trading day (Close → Open) captures the immediate market reaction once trading resumes. Second, the intraday return from the opening price of the following trading day to its closing price (Open → Close) reflects the price adjustment that occurs during the first full trading session after the announcement. Finally, the close-to-close return from the close of the earnings day to the close of the following day (Close → Close) provides a 24-hour measure of the total price reaction associated with the earnings release.

Analyzing these components separately makes it possible to distinguish between immediate repricing and subsequent price dynamics. Empirical inspection of the historical averages for the 3 time periods showed that earnings announcements often generated large absolute movements across all three measures, particularly during periods of elevated market uncertainty. Moreover, price movements during the trading session following the announcement did not consistently offset the initial overnight reaction. Instead of a systematic mean reversion, continuation of the initial move was frequently observed.

To further investigate this behavior, stocks were ranked according to their historical average absolute percentage price change around earnings announcements. This ranking made it possible to identify equities that consistently exhibited high earnings-related volatility across time. Focusing on this group of high-volatility stocks, a clear pattern emerged during 2024.

For the majority of stocks with the highest historical average absolute earnings-related price changes, earnings announcements were followed by strong directional continuation. In most cases, stocks exhibiting a positive Close → Open reaction continued with a positive Open → Close movement on the following trading day. Likewise, negative overnight reactions were frequently followed by further declines during the same session. This continuation behavior was observed for nearly all stocks belonging to this highest-volatility group, especially during periods of elevated market uncertainty.

These findings suggest that, for stocks structurally prone to large earnings-related price movements, post-earnings momentum effects can dominate short-term price dynamics. While predicting the direction of individual earnings reactions remains challenging, the systematic nature of continuation among high-volatility stocks provides a clear empirical foundation for the investment strategy developed in this project.

1.3 Identification of post-earnings momentum bias

To convert the observed post-earnings continuation behavior into a measurable and testable signal, a structured classification framework was applied to historical earnings data. The objective of this step was not to assess performance, but to identify stocks that systematically exhibit directional persistence following earnings announcements.

For each stock, every earnings event was classified according to the joint sign of the Close → Open return following the earnings announcement and the subsequent Open → Close return on the same trading day. This classification yields four mutually exclusive scenarios, distinguishing between continuation and reversal outcomes. Continuation is defined as cases where the intraday price movement has the same sign as the overnight reaction, while reversal corresponds to cases where the intraday movement offsets the initial reaction.

By applying this quadrant-based classification across all historical earnings events for each stock, empirical frequencies of continuation and reversal outcomes were computed at the individual stock level. These frequencies provide a direct measure of post-earnings momentum behavior. Figure below illustrates this classification for a single stock (AAPL), with earnings events mapped into the four quadrants according to their overnight and intraday price reactions.

```
# Post-Earnings Next-Day Outcomes for AAPL
# total_earnings=85, considered=84, excluded_c2o_zero=0, excluded_o2c_zero=1
# sum_pct=100.0
Scenario,Count,Denominator,Fraction,Percent,Dates

"After +C→O, next day UP",21,84,21/84 = 25.00%,25.0,2024-08-01;2023-05-04;2022

"After +C→O, next day DOWN",31,84,31/84 = 36.90%,36.9,2025-10-30;2025-07-31;20

"After -C→O, next day UP",16,84,16/84 = 19.05%,19.05,2024-10-31;2024-02-01;202

"After -C→O, next day DOWN",16,84,16/84 = 19.05%,19.05,2025-05-01;2023-08-03;2
```

1.4 Market-wide test: Is there a universal post-earnings momentum effect?

Before constructing the stock-selection strategy, an initial question was addressed: does post-earnings momentum represent a broad, market-wide effect across the S&P 500, or is it concentrated in a limited subset of stocks? If continuation after earnings were a general phenomenon, a uniform trading rule applied across the entire index could potentially be effective, eliminating the need for stock selection.

To examine this, the quadrants of all the stocks of the sp500 previously obtained were analyzed. Because the strategy is motivated by large informational shocks, the analysis focused specifically on high-volatility earnings events, filtering observations by the absolute size of the overnight earnings gap. Multiple thresholds were considered (|Close → Open| ≥ 5%, 7%, 8%, 9%, 10%, and 11%) in order to assess whether stronger reactions were associated with more consistent continuation behavior.

For each threshold, the relative frequency of continuation versus reversal outcomes was computed across the full universe of stocks, allowing the construction of a market-level momentum bias measure. This provided a direct test of whether continuation dominates reversal when earnings reactions are sufficiently large, independent of individual stock characteristics.

The results of this market-wide test indicate that no strong or stable continuation tendency emerges at the aggregate S&P 500 level, even when attention is restricted to large overnight earnings gaps(as can be seen in the image below, with a momentum bias of only 6.367). While certain individual stocks display persistent post-earnings continuation behavior, this effect does not generalize across the entire index. As a result, applying a single momentum rule uniformly to all S&P 500 stocks does not yield a robust or reliable edge.

```
Metric,Value
threshold_gap_pct,9
total_events,1335
count_pos_then_up,406
count_pos_then_down,342
count_neg_then_up,283
count_neg_then_down,304
pct_pos_then_up,30.412
pct_pos_then_down,25.618
pct_neg_then_up,21.1985
pct_neg_then_down,22.7715
momentum_bias_pct,6.367
```

This finding motivates the transition from a market-wide approach to a selective strategy, in which only stocks with a demonstrated historical post-earnings continuation bias are included. The next section therefore focuses on identifying and constructing a concentrated investment universe based on stock-specific momentum characteristics rather than assuming momentum is universal.

1.5 Portfolio building of the momentum strategy

Using only information available up to 2017, stocks were ranked according to the momentum bias metric defined as the difference between continuation and reversal frequencies following earnings announcements, using the quadrants previously obtained. This ranking was constructed strictly using pre-2017 data in order to avoid look-ahead bias, and no performance information from the subsequent evaluation period was used at this stage.

Based on this ranking, a Top 25 subset of stocks with the strongest historical post-earnings momentum bias was selected. To improve the statistical reliability of the signal, stocks with fewer than 40 historical earnings observations up to that point were excluded from the ranking. This constraint was introduced to reduce estimation noise and to ensure that the measured momentum bias was based on a sufficiently large number of earnings events.

```
rank,ticker,considered,total_earnings,pos_then_up,pos_then_down,neg_then_up,neg_then_down,continuation,reversal,momentum_bias
1,FSLR,44,44,36.36363636363637,11.363636363636363,9.090909090909092,43.18181818181818,79.54545454545455,20.454545454545453,59.0
2,CMG,46,46,30.434782608695652,8.695652173913043,19.565217391304348,41.30434782608695,71.73913043478261,28.26086956521739,43.47
3,VZ,45,45,48.888888888888886,20.0,8.88888888888889,22.22222222222222,71.11111111111111,28.88888888888889,42.22222222222223
4,WY,45,45,28.88888888888889,20.0,8.88888888888889,42.22222222222222,71.11111111111111,28.88888888888889,42.22222222222223
5,RSG,54,54,40.74074074074074,22.22222222222222,7.407407407407407,29.62962962962963,70.37037037037037,29.62962962962963,40.7407
6,ISRG,54,54,42.592592592592595,11.11111111111111,20.37037037037037,25.925925925925927,68.51851851851852,31.48148148148148,37.0
7,SMCI,41,41,36.58536585365854,24.390243902439025,7.317073170731708,31.70731707317073,68.29268292682927,31.70731707317073,36.58
8,MSCI,40,40,55.0,15.0,17.5,12.5,67.5,32.5,35.0
9,EXPE,49,49,36.734693877551024,16.3265306122449,16.3265306122449,30.612244897959183,67.34693877551021,32.6530612244898,34.6938
10,MOH,48,48,39.583333333333336,16.666666666666668,16.666666666666668,27.083333333333332,66.66666666666667,33.333333333333336,3
11,FE,48,48,31.25,16.666666666666668,16.666666666666668,35.416666666666664,66.66666666666666,33.333333333333336,33.333333333333
12,BXP,47,47,36.170212765957444,17.02127659574468,17.02127659574468,29.78723404255319,65.95744680851064,34.04255319148936,31.91
13,EW,49,49,32.6530612244898,14.285714285714286,20.408163265306122,32.6530612244898,65.3061224489796,34.69387755102041,30.61224
14,SYY,49,49,30.612244897959183,16.3265306122449,18.367346938775512,34.69387755102041,65.3061224489796,34.69387755102041,30.612
```

This strict separation between signal identification and performance evaluation represents a central methodological feature of the analysis. By defining the stock selection exclusively on pre-2017 data and evaluating performance only in the subsequent period, the analysis ensures that observed results reflect out-of-sample behavior rather than in-sample optimization or data-snooping effects.

All stocks included in the Top 25 subset were subsequently used as the investment universe for the out-of-sample trading strategy. The rules applied to this portfolio strategy, as well as the resulting performance, are described and analyzed in the following sections in order to assess whether the strategy yields meaningful returns.

2. Analysis and selection of performance metrics

2.1 Trading strategy definition

The trading strategy analyzed in this project is an event-driven, post-earnings momentum strategy applied to the 25 fixed universe of stocks selected before the evaluation period. The strategy operates exclusively around corporate earnings announcements and does not rely on continuous market exposure.

For each stock in the Top 25 momentum-biased universe, earnings announcements are released after the close of the trading session on the earnings day. As a result, the initial market reaction to the earnings information is reflected in the price movement from the close of the earnings day to the opening price of the following trading day (Close → Open).

The core trading rule is defined as follows. At the market open of the trading day following the earnings release, the Close → Open return is observed and used to determine position direction. If the overnight return is positive, a long position is entered at the market open. If the overnight return is negative, a short position is entered. In both cases, the position is closed at the end of the same trading session. The realized return of each trade corresponds to the intraday price movement from open to close (Open → Close), adjusted for the direction of the position.

The strategy does not attempt to predict earnings surprises or price direction in advance. Instead, it relies on the historically observed momentum bias of the selected stocks and takes positions with the expectation that post-earnings price movements will tend to follow their historical continuation patterns.

Given that the investment universe consists of 25 stocks and that each stock reports earnings approximately four times per year, the strategy generates roughly 100 trades per year. On trading days when no earnings-related positions are opened, the portfolio capital is assumed to remain fully in cash. This cash position does not generate returns, but it also does not incur losses. The out-of-sample trading period considered in the analysis spans from the beginning of 2017 to the end of 2024.

2.2 Portfolio construction and return computation

Portfolio performance is computed at the daily level by aggregating all earnings-driven trades executed on the same trading day and compounding the resulting returns over time. Because the strategy is activated only around earnings announcements, trading activity is intermittent rather than continuous.

On any given trading day, the number of executed trades depends on how many stocks in the Top 25 portfolio released earnings the previous day. When no earnings-related trades occur, the portfolio remains fully invested in cash. In these cases, the daily portfolio return is assumed to be zero, meaning that capital neither increases nor decreases, as explained in the previous section.

When one or more trades occur on the same day, capital is allocated equally across all active positions. This equal-weighting is implemented by computing the arithmetic average of the individual trade returns generated on that day. Each trade return is already adjusted for position direction, meaning that long positions use the Open → Close return directly, while short positions use the negative of that return.

The daily portfolio return is therefore computed as:

N = number of trades executed on day t

r_i = return of trade i on day t (direction-adjusted)

daily_portfolio_return_t = (1 / N) * (r_1 + r_2 + ... + r_N)

If no trades occur on day `t`, the return is defined as:

daily_portfolio_return_t = 0

After the daily portfolio return has been computed, portfolio equity is updated through compounding. This means that gains and losses affect the capital base used for subsequent trading days, rather than being added linearly. The equity update rule is:

equity_after_t =

equity_before_t * (1 + daily_portfolio_return_t)

This compounding process is applied sequentially for each trading day in the evaluation period, resulting in a cumulative equity curve that reflects the full effect of the strategy over time.

To illustrate this mechanism, consider a simple numeric example. Suppose the portfolio equity at the start of a trading day is normalized to 1.00. On that day, three stocks generate earnings-driven trades with the following returns:

returns = [0.02, -0.01, 0.04]

The daily portfolio return is the average of these values, since the total capital was divided equally before each trade:

daily_portfolio_return =

(0.02 - 0.01 + 0.04) / 3 = 0.016667

Equity is then updated by compounding:

equity_after =

1.00 * (1 + 0.016667) = 1.016667

If the following trading day contains no earnings announcements, the portfolio remains in cash:

daily_portfolio_return = 0

equity_after = 1.016667

If a subsequent day contains a single trade with a return of 3%, the update becomes:

daily_portfolio_return = 0.03

equity_after = 1.016667 * (1 + 0.03) = 1.047167

This example illustrates two important features of the portfolio construction. Exposure is evenly distributed across all trades executed on the same day, limiting concentration risk on high-activity earnings dates. In addition, performance is driven by compounding daily returns, so the timing and sequence of gains and losses have a meaningful effect on long-term outcomes.

2.3 Selection of performance metrics

Once the trading strategy and portfolio construction rules have been defined, the next step is to determine how the performance of the strategy should be evaluated. The choice of output variables is a critical component of the analysis, as no single metric is sufficient to fully characterize the behavior of an investment strategy. Different outputs capture different dimensions of performance, such as long-term growth, variability, concentration of returns, and sensitivity to timing.

The outputs selected in this project were chosen with the explicit objective of enabling a transparent and structured comparison between the proposed strategy and a passive benchmark investment in the S&P 500 over the same period. Together, they provide complementary perspectives on return generation and risk exposure, rather than a single-dimensional assessment, and gives the answer of whether active trading offers a meaningful advantage over passive index investment.

The primary output is the **equity curve**, which represents the cumulative evolution of portfolio value over time under continuous compounding. By rebasing both the strategy and the benchmark to a common starting value, the equity curve allows for a direct comparison of performance paths rather than only final outcomes. This output makes it possible to observe differences in growth dynamics, drawdowns, and the temporal distribution of returns, highlighting whether performance is smooth or concentrated in specific periods.

From the equity curve, the **compound annual growth rate (CAGR)** is computed for both the strategy and the benchmark. CAGR provides a standardized annualized measure of long-term performance, allowing for an intuitive comparison across investments with the same horizon. While CAGR does not capture volatility or path dependence, it serves as a widely accepted summary statistic for overall growth and is therefore included as a reference measure.

In addition to the full-period CAGR, **forward-looking CAGRs** are computed for every possible starting month within the evaluation window. For each month, CAGR is calculated from the beginning of that month until the end of the sample period. This rolling-start approach allows the analysis to assess whether the strategy would still have outperformed the S&P 500 depending on the entry point. By comparing the forward CAGR of the strategy and the benchmark for each starting month, this output provides insight into the sensitivity of long-term performance to timing and helps evaluate whether outperformance is robust across most entry dates or concentrated in a limited subset of months.

To complement long-term aggregates, returns are also analyzed at lower frequencies. **Monthly returns** are computed in order to assess the consistency of performance across time. Examining returns at the monthly level helps identify whether performance is driven by a small number of extreme observations or whether gains are more evenly distributed. This perspective is particularly relevant for an event-driven strategy, where returns may cluster around specific dates.

In addition, **yearly returns** are computed to evaluate performance across different market regimes .

Another graphic produced is one that illustrates the **percentage evolution for each month** of the strategy compared to that same month in the sp500, to help later explain the main causes of the strong and poor months in the period.

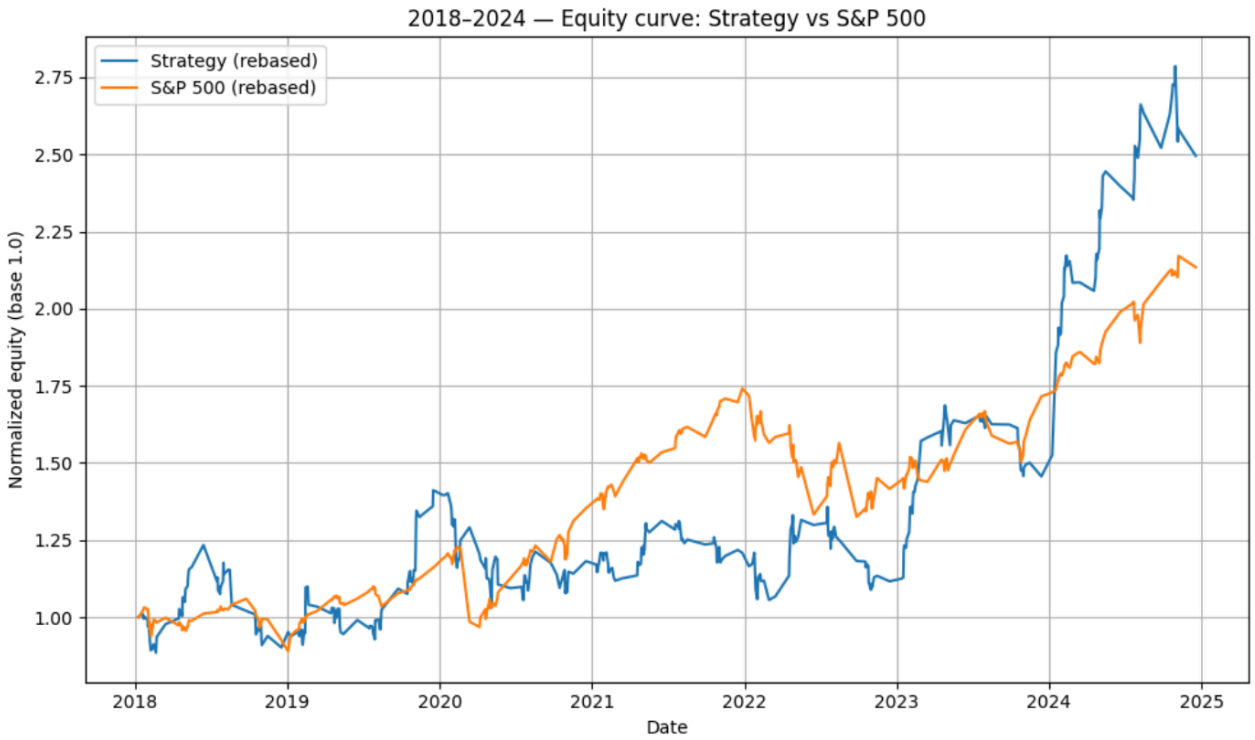
Finally, **return distributions** are analyzed using bell-shaped density plots. These distributions allow for a comparison of the dispersion, skewness, and concentration of returns between the strategy and the benchmark. By examining the shape of the distributions rather than only their central tendency, it becomes possible to assess whether performance is driven by frequent moderate returns or by rare but extreme outcomes.

All this metrics will be shown and analyzed in the following section.

3. Results analysis

3.1 Equity curve and long-term performance

The equity curve provides the most direct and intuitive representation of how the trading strategy performs over time relative to the benchmark. By rebasing both the strategy and the S&P 500 to a common starting value of 1.0 at the beginning of 2018, the visualization allows for a direct comparison of the growth paths that one unit of invested capital would have followed under each approach. Unlike summary statistics that focus only on final outcomes, the equity curve reveals the full investment trajectory, including periods of acceleration, stagnation, drawdown, and recovery.



A first inspection of the equity curves shows that both the momentum strategy and the S&P 500 generate positive long-term growth over the evaluation period from 2018 to 2024. However, the manner in which this growth is achieved differs substantially. The S&P 500 follows a relatively smooth and gradual upward trajectory, consistent with the behavior of a broadly diversified index with continuous market exposure. In contrast, the strategy's equity curve exhibits stronger fluctuations, characterized by sharper upward movements during certain intervals and flatter or declining phases during others.

These differences become particularly evident during periods of market stress and elevated volatility. For example, during the COVID-19 shock in March 2020, both the strategy and the benchmark experienced a pronounced drawdown, reflecting the systemic nature of the

market sell-off. However, in subsequent high-volatility environments—most notably during 2019 and again throughout 2024—the strategy exhibits strong relative outperformance compared with the S&P 500. These episodes coincide with periods in which earnings-related price movements intensified, creating conditions under which the event-driven momentum mechanism of the strategy appears to be more effective.

This contrast highlights an important structural distinction between the two approaches. While both investments are ultimately exposed to broad market dynamics, the strategy responds differently to volatility regimes. Its performance is less uniform over time, but it shows the ability to capitalize on environments characterized by sharp repricing and increased dispersion across individual stocks. As a result, returns tend to cluster in specific periods rather than being evenly distributed across the sample.

To summarize long-term performance in a standardized and comparable way, the compound annual growth rate (CAGR) is computed for both the strategy and the benchmark over the full sample period. CAGR represents the constant annual growth rate that would transform the initial portfolio value into the final value over the given horizon, abstracting from intermediate fluctuations.

Over the 2018–2024 period, the strategy achieves a CAGR of 14.082 % per year, while the S&P 500 achieves a CAGR of 11.543% per year, resulting in a difference of +2.539 percentage points per year in favor of the strategy. At a purely aggregate level, this indicates that the strategy delivers higher annualized growth than the benchmark over the evaluation window.

However, the equity curve makes clear that this higher CAGR is not achieved through a smoother or more stable path. Instead, it results from periods of strong relative outperformance that compensate for extended intervals of weaker or flat relative performance. This illustrates a key limitation of CAGR as a standalone metric: while it summarizes long-term growth effectively, it does not convey information about the timing, variability, or risk profile of returns.

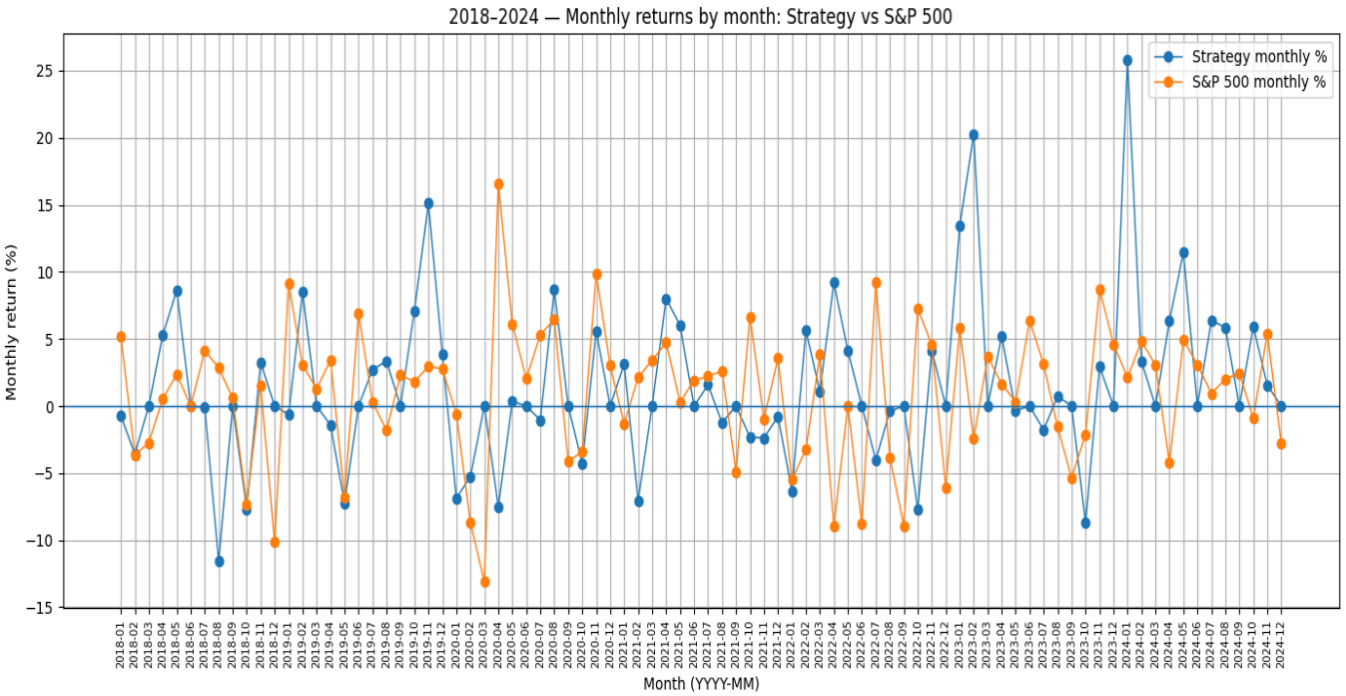
An additional observation from the equity curves concerns capital preservation relative to the initial investment. When the portfolio value is normalized to 1.0 at the start of 2018, the S&P 500 temporarily falls below this initial level during several periods, most notably during the sharp market drawdown of March 2020. In contrast, following the 2020 market shock, the momentum strategy does not fall below the initial investment level over the remainder of the sample period. However, throughout 2021 and 2022 its cumulative value remains below that of the S&P 500, indicating that the initial investment grows less over this interval despite not implying inferior annual performance in each individual year. Overall, drawdowns in the strategy are intermittent and concentrated around specific episodes rather than reflecting continuous exposure to broad market declines

This difference should not be interpreted as evidence of structural downside protection. Rather, it reflects the specific exposure profile of the strategy during the sample period. Because the strategy is active only on earnings-related trading days and remains in cash otherwise, its exposure to broad market sell-offs is intermittent rather than continuous. As a

result, certain market drawdowns affect the benchmark more directly than the strategy. It is also important to acknowledge that this outcome may partially reflect favorable timing within the evaluation window rather than a guaranteed property of the strategy.

Overall, the equity curve analysis indicates that long-term outperformance, while present in aggregate terms, is achieved in a non-uniform manner and depends critically on the timing of return realization. This path dependence raises an important practical question: to what extent does the observed outperformance depend on the specific entry point chosen by the investor? This question is addressed explicitly in the following section through a forward-looking CAGR analysis that examines sensitivity to entry timing across different starting months.

While the equity curve summarizes cumulative performance, a monthly return representation helps isolate the specific timing and concentration of return contributions, that's why a monthly returns plot is also included. This visualization highlights how performance differences emerge at specific points in time rather than being evenly distributed. For example, although the S&P 500 recovered rapidly from the COVID-19 shock in March 2020, the strategy experienced an additional decline in April 2020, reflecting its sensitivity to post-earnings dynamics during periods of extreme uncertainty.



Similarly, during the geopolitical shock associated with the war in Ukraine in 2022, both the strategy and the S&P 500 experienced negative performance, indicating that broad market stress affected both approaches. However, the magnitude and timing of losses differ, reflecting the strategy's intermittent exposure. In contrast, certain months stand out as periods of pronounced relative outperformance for the strategy. In particular, November 2019 and January 2024 show substantial positive deviations relative to the S&P 500, corresponding to environments of elevated volatility in which the post-earnings momentum mechanism appears to be especially effective.

Overall, this monthly perspective reinforces the conclusion that the strategy's performance is driven by a limited number of critical periods. Long-term outperformance is therefore the result of clustered gains during specific months rather than consistent superiority across the entire sample.

3.2 Forward-looking performance and sensitivity to entry timing

While the equity curve and full-period CAGR provide valuable insight into long-term performance, they implicitly assume a single entry point at the beginning of the evaluation period. In practice, however, investors do not enter a strategy at a unique or predetermined date. The realized performance of an investment strategy can depend significantly on the timing of entry, particularly for strategies whose returns are concentrated in specific market regimes or clustered around certain events.

To assess this sensitivity, a forward-looking performance analysis is conducted. Rather than computing a single CAGR over the full sample, the strategy's compound annual growth rate is recalculated multiple times using different starting points. Specifically, for each calendar month within the evaluation period, CAGR is computed from the beginning of that month until the end of the sample in December 2024. The same procedure is applied to the S&P 500 benchmark.

Formally, for each starting month t, the forward-looking CAGR is defined as:

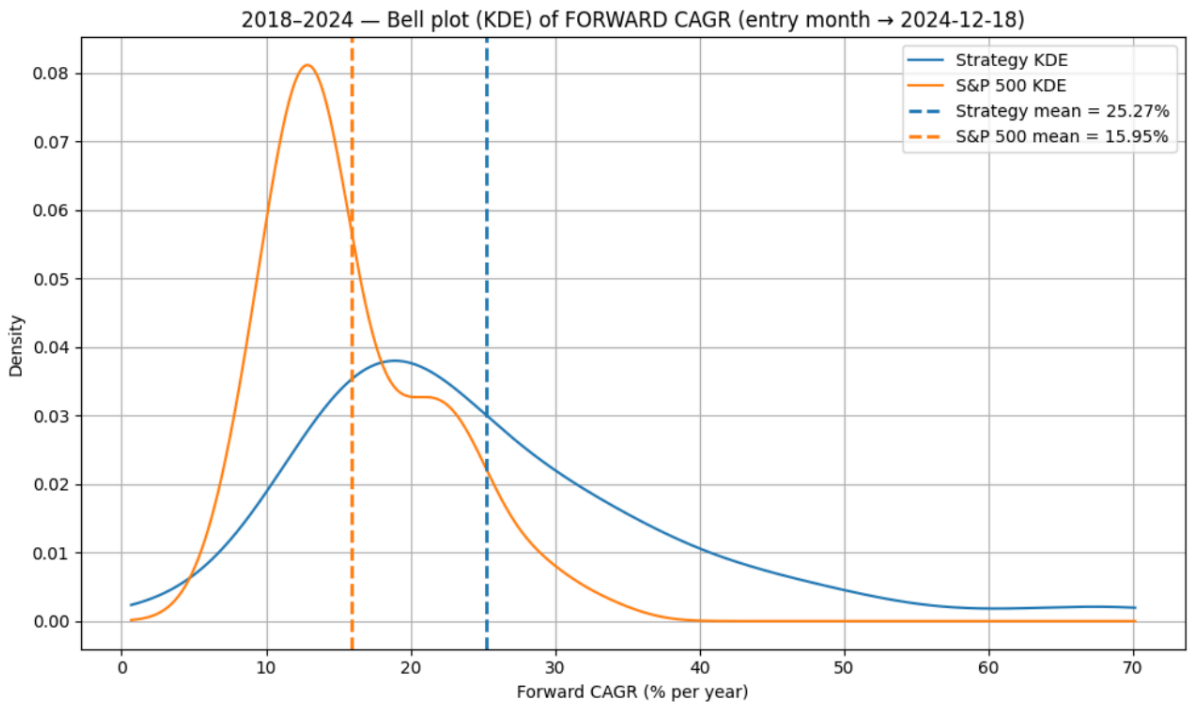
$$CAGR_t = (Equity_{end} / Equity_t)^{(1 / Years_{remaining})} - 1$$

where: $Equity_t$ is the portfolio value at the beginning of month t,
$Equity_{end}$ is the portfolio value at the end of December 2024,
$Years_{remaining}$ is the fraction of years between t and the end of the sample.

This rolling-start approach produces a distribution of forward CAGRs rather than a single aggregate figure. Each observation answers the question: "If an investor had started allocating capital to this strategy in month t and held it until the end of the sample, how would the annualized return compare to the benchmark?"

Overall, the forward-looking CAGR analysis provides a direct measure of robustness and path dependence. By evaluating performance across all possible entry months, it reveals whether long-term outperformance is broadly accessible or concentrated in specific regimes. For an event-driven strategy, this distinction is crucial, as returns may cluster during periods of elevated volatility rather than uniformly over time.

As illustrated in the figure below, each observation in the forward CAGR distribution corresponds to a distinct entry month and represents the annualized return an investor would have obtained by entering the strategy at that point and holding the position until the end of the sample period.



The comparison between the strategy's forward CAGRs and those of the S&P 500 helps assess whether the observed outperformance is broadly accessible or strongly dependent on entry timing. By computing forward-looking CAGRs for all possible entry months, it becomes possible to evaluate how often an investor would have achieved higher long-term returns by allocating capital to the strategy rather than to the benchmark.

The results show that in **88.8% of all possible entry months**, the strategy delivers a higher forward CAGR than the S&P 500 when measured until the end of 2024. This outcome does not imply that the strategy dominates the benchmark in a structural or unconditional sense. Instead, it reflects the combined effect of strong performance in the later part of the sample and the fact that forward CAGRs are measured relative to a common end date. As a result, entry points closer to periods of strong strategy performance mechanically benefit from a longer exposure to favorable conditions.

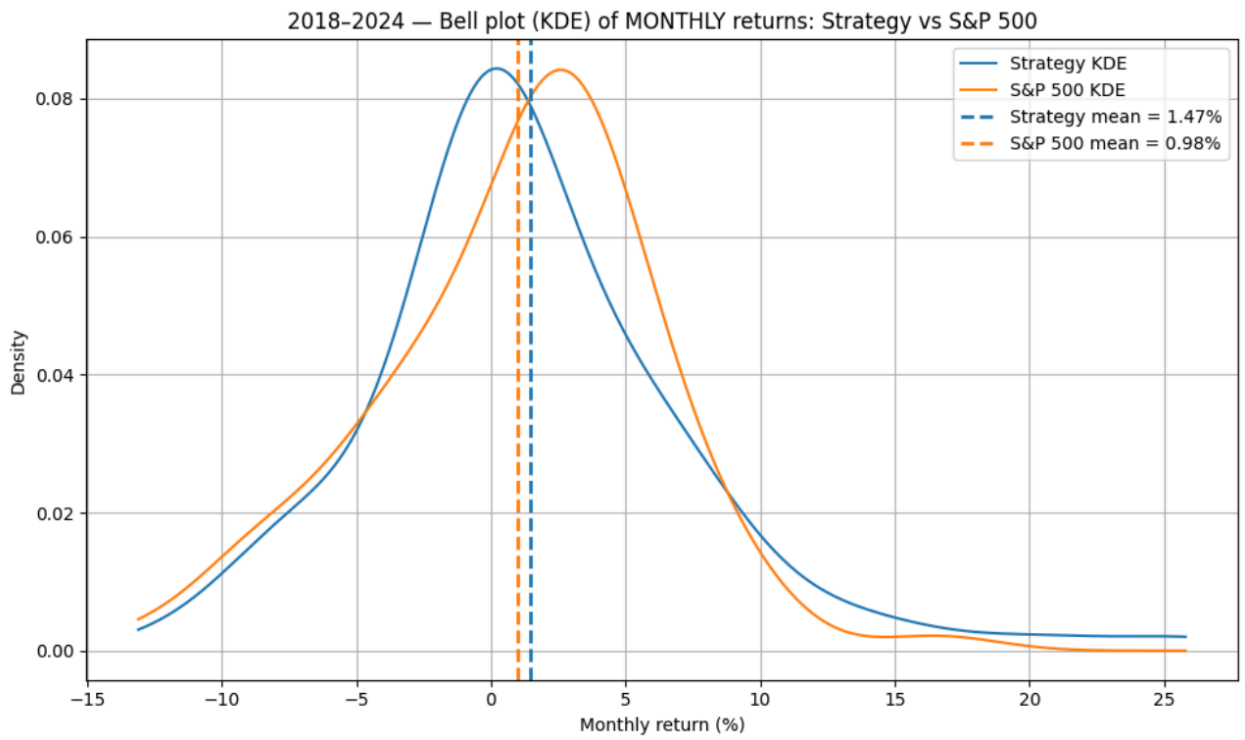
This interpretation is supported by the forward CAGR distributions shown in the figure. The strategy's distribution is shifted to the right relative to that of the S&P 500, indicating higher average long-term outcomes, but it is also noticeably wider. This wider spread reflects greater sensitivity to entry timing: while many entry months lead to very strong long-term performance, others result in substantially lower outcomes. In contrast, the S&P 500 exhibits a more concentrated distribution, consistent with a smoother but less convex return profile.

Importantly, this high fraction of favorable entry months does not contradict the fact that the strategy outperforms the S&P 500 in only 47.6% of individual months, as forward CAGRs aggregate long-horizon effects rather than month-by-month performance.

Overall, the forward-looking analysis suggests that the strategy offers the potential for significantly higher long-term returns than the benchmark, but at the cost of increased path dependence. Long-term outperformance is therefore not the result of consistent superiority across all periods, but rather of stronger performance in specific regimes that disproportionately influence long-horizon outcomes.

3.3 Distribution of monthly returns

To complement the analysis of cumulative performance, CAGR, and sensitivity to entry timing, this section examines the distribution of monthly returns for both the momentum strategy and the S&P 500. Rather than focusing on aggregated growth or specific entry points, this perspective analyzes how returns are distributed at the monthly level, providing insight into variability, tail behavior, and the underlying risk characteristics of each approach. The figure below presents kernel density estimates (KDEs) of monthly returns from 2018 to 2024 for the strategy and the benchmark. These bell-shaped curves represent smoothed empirical distributions of observed monthly returns and do not assume that returns follow a normal (Gaussian) distribution. Instead, they summarize the relative frequency of different outcomes over the sample period.



A first observation is that the peak (mode) of the S&P 500 distribution is positioned slightly further to the right than that of the strategy, indicating that the most common monthly return for the benchmark is marginally higher.
However, this does not indicate that the benchmark constitutes a superior strategy. Although the most frequent monthly return of the S&P 500 is marginally higher, the mean of the strategy's return distribution lies to the right of that of the benchmark, indicating a higher average monthly return despite a lower most-frequent outcome.

The key distinction lies in the shape and dispersion of the distributions. The momentum strategy's return distribution is visibly wider, with heavier tails on both the positive and negative sides. On the right tail, the strategy exhibits a higher probability of very large positive monthly returns (exceeding +10%), consistent with the sharp outperformance episodes identified in earlier sections. These extreme positive outcomes are largely absent from the S&P 500 distribution.
On the left tail, the strategy also displays a greater likelihood of the most negative months, reflecting higher downside risk relative to the benchmark, although the density in the intermediate range between –5% and –10% is more or less the same as observed for the S&P 500.
This asymmetric widening of the distribution captures the fundamental trade-off underlying the momentum strategy. While it does not maximize the most common monthly outcome, it increases exposure to high-magnitude return realizations. As a result, performance becomes more sensitive to timing: entering the strategy before favorable clusters of events can lead to substantial outperformance, whereas entering before unfavorable periods can result in pronounced drawdowns.

In contrast, the S&P 500 distribution is more concentrated around its mean, with thinner tails. This reflects a smoother return profile and lower sensitivity to individual months, which reduces timing risk but also limits upside convexity. The KDE comparison therefore reinforces the conclusions drawn from the equity curve and forward-looking CAGR analyses: the momentum strategy offers higher potential upside at the cost of increased volatility and stronger path dependence, whereas the benchmark delivers more stable but less extreme outcomes.

3.4 Distribution of yearly returns

In addition to monthly return behavior, performance is also examined at the yearly level, as annual returns are the standard horizon used in portfolio reporting and investment evaluation. Analyzing yearly outcomes allows short-term volatility to be aggregated into economically meaningful results that better reflect how investors experience performance over time.

Figure below shows kernel density estimates (KDEs) of calendar-year returns for the momentum strategy and the S&P 500 over the 2018–2024 period. These distributions summarize the dispersion and tail behavior of annual outcomes without imposing any parametric assumptions.

At this horizon, the S&P 500 exhibits a relatively concentrated distribution, with most annual returns clustered around moderately positive values. This reflects stable year-to-year performance driven by continuous market exposure and broad diversification. Extreme outcomes, both positive and negative, are comparatively rare.

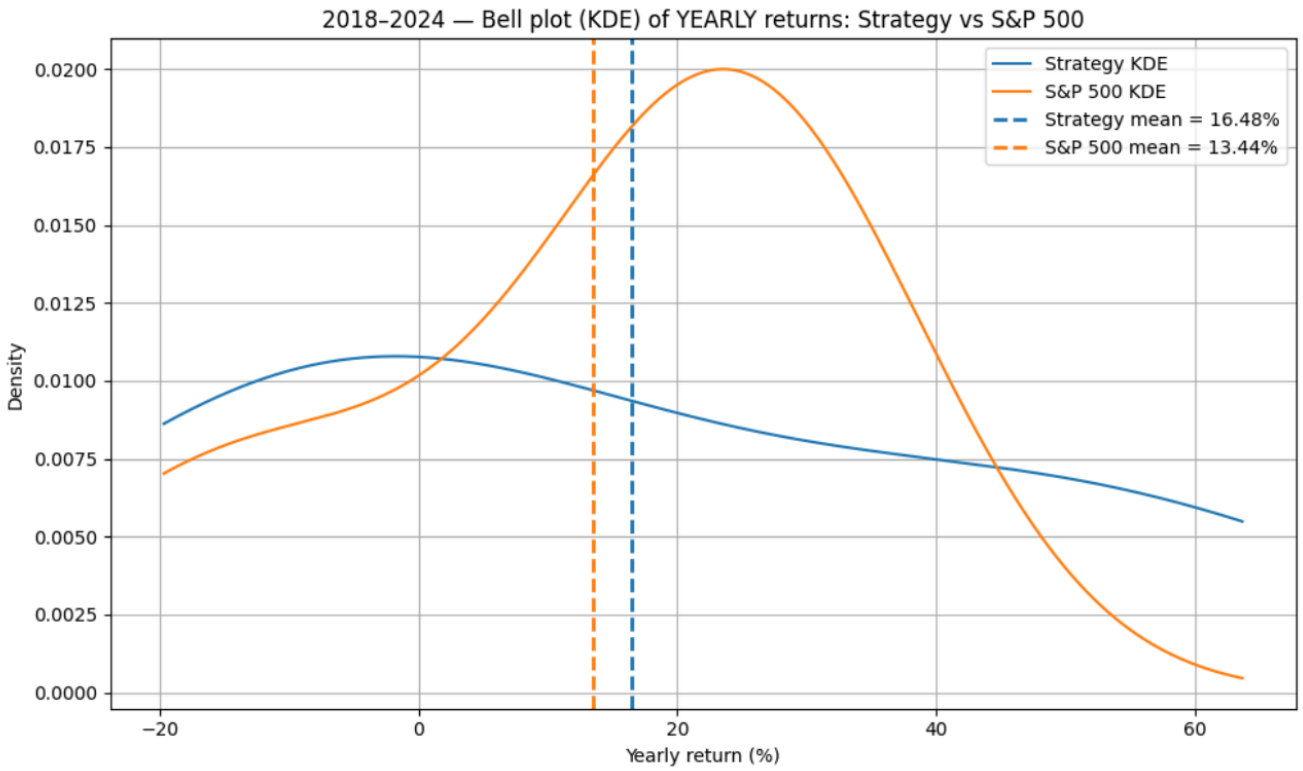
The momentum strategy displays a markedly wider yearly return distribution. While its mean annual return is higher, this higher average is accompanied by substantially greater dispersion. The right tail extends to much larger positive annual returns, indicating that a significant portion of total performance is generated in a small number of exceptionally strong years. Conversely, the left tail is also wider, signaling higher downside risk in weaker years.

In this analysis, in the years 2020,2021 and 2018 the sp500 performed better than the strategy, while in the rest of years 2019,2022,2023,2024, the strategy overperformed the sp500, specially in 2019 and 2024, where the dropdowns of the bad years are completely compensated as can be seen in here:

```
--------------------------------------------------------------------------------
2018-2024 — Yearly returns (Strategy vs S&P 500)
--------------------------------------------------------------------------------
year, strategy_yearly_%, sp500_yearly_%, diff_pp
2018, -9.823, -6.591, -3.232
2019,  48.275,  30.433,  17.842
2020, -15.344,  15.761, -31.106
2021,  3.243,  26.605, -23.362
2022, -4.151, -19.644,  15.494
2023,  29.531,  23.786,  5.745
2024,  63.604,  23.749,  39.855

SUMMARY (yearly):
% years Strategy > S&P 500: 57.1%
Mean Strategy yearly return: 16.476%
Mean S&P 500 yearly return:  13.443%
```

This pattern suggests that the strategy's long-term performance is not driven by consistently strong annual returns, but by a small number of large outcomes. Looking at yearly results alongside the monthly and forward-looking analyses helps clarify how short-term volatility translates into uneven performance from year to year. Overall, the momentum strategy offers higher upside potential, but at the cost of greater variability and stronger path dependence, whereas the S&P 500 exhibits more stable, but with lower upside potential, annual returns.



4.Pipeline execution

The analysis is executed as a linear and reproducible pipeline of Python scripts, where each stage produces structured outputs that serve as inputs for the subsequent step. The workflow begins with the construction of post-earnings price reaction classifications at the individual stock level and progressively aggregates, filters, and evaluates these signals to produce the final strategy results.

First, **run_quadrants.py** applies a post-earnings quadrant classification to each stock in the S&P 500. The classification logic is implemented in the internal module quadrants.py, where each earnings event is decomposed into overnight (Close → Open) and intraday (Open → Close) returns and assigned to one of four mutually exclusive continuation or reversal categories based on the sign of these returns. The output consists of detailed per-stock files containing all earnings-level observations and their associated quadrant labels.

Next, **top_25_quadrants_until2017.py** constructs the investment universe using only information available up to the end of 2017. Based on the historical quadrant frequencies, stocks are ranked according to their post-earnings momentum bias, defined as the difference between continuation and reversal frequencies. To ensure statistical reliability, stocks with insufficient earnings observations are excluded, and the top 25 stocks are selected as the fixed universe for subsequent out-of-sample evaluation.

The trading strategy is then executed using **momentum_portfolio_top25_until_2024.py**. This script applies the predefined trading rule to the selected stock universe over the out-of-sample period, simulating earnings-driven trades on a day-by-day basis. When multiple earnings events occur on the same day, capital is allocated equally across all active positions, and portfolio returns are compounded over time to generate a complete equity time series.

Finally, **performance_analysis_and_plots.py** evaluates strategy performance and produces all figures and summary statistics used in the analysis. This includes the construction of equity curves, the computation of monthly and yearly returns, forward-looking CAGR distributions across different entry points, and direct comparisons with a passive S&P 500 benchmark.

run_quadrants.py

↓

top_25_quadrants_until2017.py

↓

momentum_portfolio_top25_until_2024.py

↓

performance_analysis_and_plots.py

5. Conclusions, risks, and potential improvements

This project set out to evaluate whether post-earnings price dynamics can be exploited through an event-driven momentum strategy using a rigorous out-of-sample backtesting framework. After identifying a subset of stocks exhibiting persistent post-earnings continuation behavior, the strategy was defined and tested on a fixed universe of 25 stocks selected exclusively using pre-2018 information. The subsequent backtest, conducted over the period from 2018 to December 2024, demonstrates that earnings-related momentum can generate economically meaningful returns when evaluated under realistic trading constraints and strict avoidance of look-ahead bias.

Across the 2018–2024 evaluation period, the strategy achieves higher long-term growth than the S&P 500, as measured by CAGR, while exhibiting substantially different performance dynamics. Rather than delivering smooth, index-like returns, performance is concentrated in specific periods characterized by elevated volatility and strong earnings-driven repricing. The equity curve, forward-looking CAGR analysis, and return distributions consistently show that outperformance is achieved through clustered gains rather than uniform superiority across time.

At the same time, the results highlight important risks and limitations. The strategy displays higher volatility, stronger path dependence, and greater sensitivity to entry timing than a passive benchmark. While the strategy avoids continuous exposure to market drawdowns by remaining in cash outside earnings windows, it is not inherently protected against losses. Periods of weak performance and drawdowns do occur, and favorable outcomes over the sample period may partially reflect the specific volatility regimes observed rather than a universal structural edge.

In addition, the forward-looking analysis shows that a large fraction of entry months benefit from these clustered gains when measured relative to a common end date, reinforcing the importance of horizon choice and timing in interpreting long-term performance.

A key limitation of the current implementation lies in its binary and unconditional trading rule. Positions are taken solely based on the sign of the overnight price reaction, implicitly assuming that continuation probabilities are symmetric across directions. However, the empirical analysis suggests that this assumption does not always hold. For certain stocks, continuation may be significantly stronger in one direction than the other. For example, a stock may exhibit a high probability of continued downside following negative earnings reactions, while positive reactions revert more frequently. This asymmetric behavior is not fully captured by the aggregate continuation metric used to select the Top 25 universe.

An important extension of the strategy would therefore be to condition trade direction on the dominant historical continuation bias, rather than always following the observed overnight move. Instead of systematically trading long after positive reactions and short after negative ones, position direction could be filtered or weighted based on the stock-specific probability of continuation in each quadrant. Incorporating such conditional logic could increase the statistical edge of the strategy, reduce unnecessary exposure in weak regimes, and potentially improve the results.

More broadly, the results suggest that post-earnings momentum strategies are best understood as volatility-responsive mechanisms. When applied carefully and combined with stronger signal conditioning, such strategies may offer exposure to “high-impact” return opportunities—what could be described as rocket-like returns—while maintaining controlled exposure through limited trading frequency and cash positioning outside events.

In conclusion, this study demonstrates that earnings-driven momentum effects are real, measurable, and exploitable under certain conditions, but also that their effective use requires careful attention to timing, conditioning, and risk management. With further refinement, particularly through asymmetric signal selection and adaptive filtering, this framework could serve as a foundation for more robust event-driven trading strategies that balance high upside potential with disciplined risk control.

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Github Repository:

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